

Optimal Credit Limit Extension Policy

Using Multi-Period Lifecycle Simulation

A Risk-Adjusted Profitability Framework for Consumer Lending

Praveen Kumar, praveenkm31@gmail.com, +918585010275
<https://www.linkedin.com/in/praveenk-8/>

30,000 Accounts · 12 Monthly Campaigns · Monte Carlo + LP
Optimisation

Executive Summary

The opportunity in numbers — 12-month lifecycle simulation

\$1.41M

Net Profit
(66.6% margin)

0.66%

Portfolio Default Rate
(cap: 5%)

70,317

Extensions Granted
across 12 campaigns

96.1%

Customers with
≥1 Offer Accepted

81.8%

Good→Excellent
Migration Rate

Risk Well Within Tolerance

Portfolio default rate of 0.66% sits 7.5× below the 5% LP risk cap. Peak campaign-level exposure never exceeded 1.0% across all 12 campaigns.

Credit Quality Self-Improves

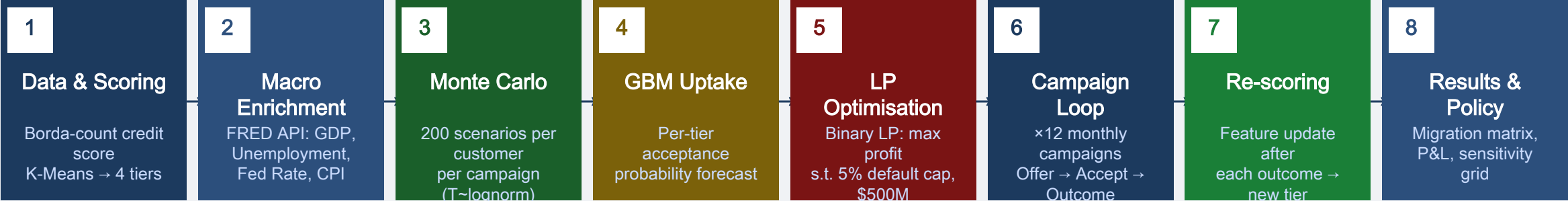
83.8% of Fair-tier customers improved to Good or Excellent by year-end. The re-scoring mechanism creates compounding credit quality gains at no additional cost.

LP Optimisation is Conservative

The \$500M exposure cap was never approached. Risk headroom exists to expand offer volumes to Good and Excellent tiers without constraint violations.

Methodology

Eight-stage pipeline from raw data to actionable policy



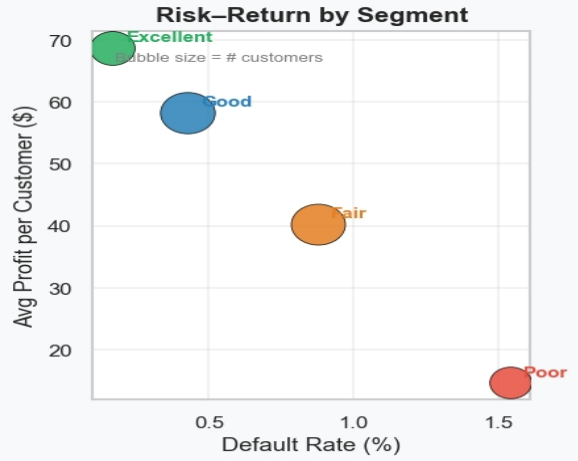
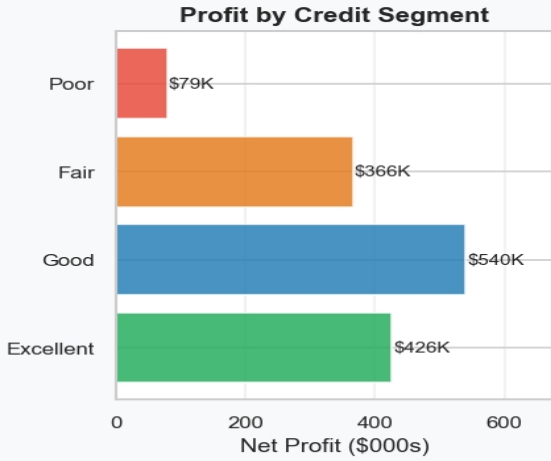
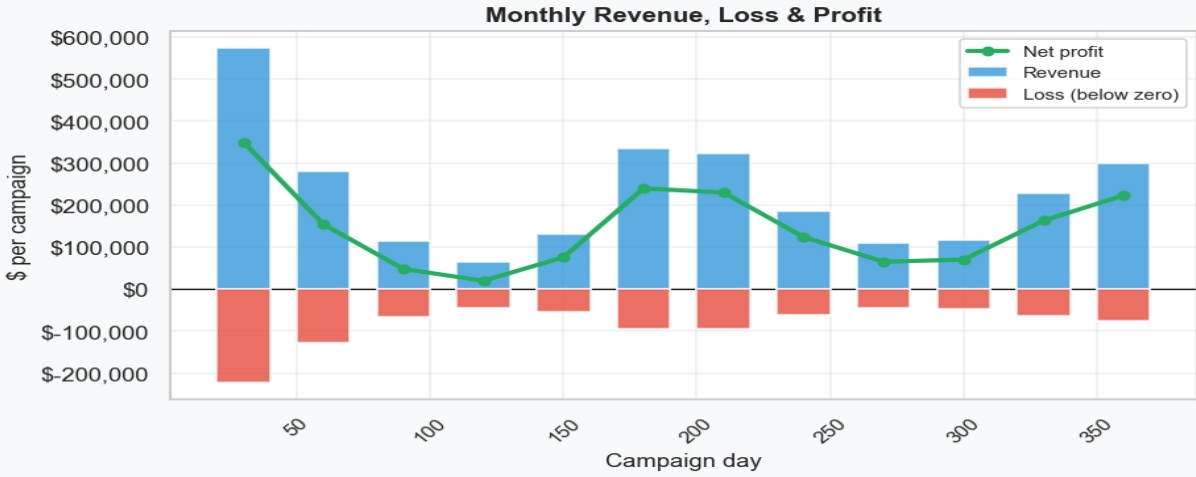
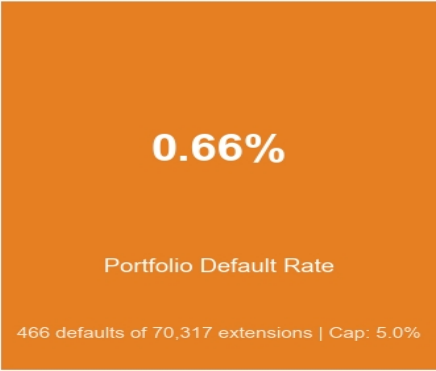
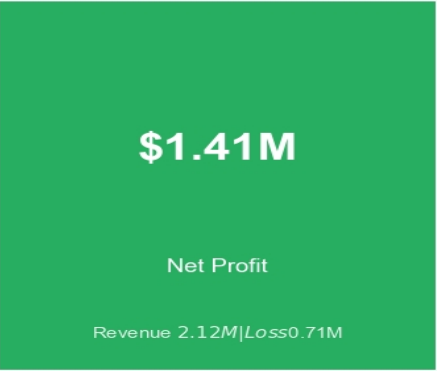
Key Differentiator vs One-Shot Models

- Explicit Bernoulli acceptance draw per campaign (not implicit uptake weighting)
- Actual lognormal term draw at moment of acceptance — not Monte Carlo mean
- Re-scoring after every outcome: on-time +2pp, early +1pp, default −10pp
- Credit state evolves dynamically — LP sees the updated risk each month

12-Month Portfolio Performance

Overview dashboard — monthly P&L, volume, state drift, risk

Credit Limit Extension — Executive Dashboard



Performance by Credit Tier

Each tier tells a different risk-return story

Tier	Customers	Accept Rate	Avg Profit / Cust	Default Rate	Margin	Year-End Migration
Excellent	6,206	79.5%	\$68.70	0.17%	90.8%	99.5% remain Excellent
Good	9,267	74.3%	\$58.25	0.43%	76.5%	81.8% → Excellent
Fair	9,089	61.8%	\$40.23	0.88%	55.4%	83.8% → Good / Excellent
Poor	5,438	44.3%	\$14.62	1.54%	28.0%	30.6% → Good
Portfolio	30,000	65.1%	\$47.04	0.66%	66.6%	—

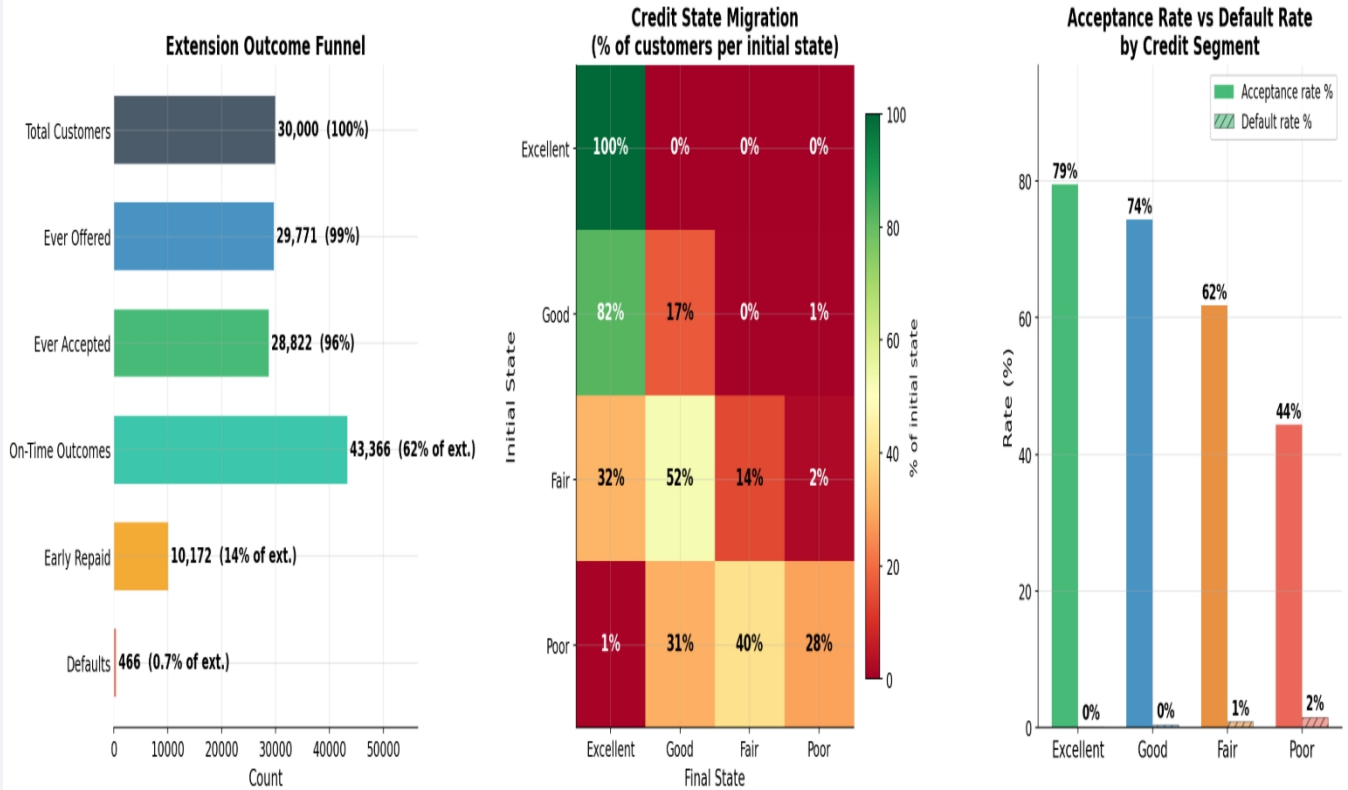
Key Insight: Portfolio Risk Budget is Vastly Under-Utilised

- The 0.66% portfolio default rate is 7.5× below the 5% LP constraint — the optimiser is inherently conservative.
- The 9× gap between Excellent (0.17%) and Poor (1.54%) default rates validates the four-tier segmentation as meaningful risk differentiation.
- Good-tier accounts drive the largest absolute profit (\$539K) despite lower per-customer yield than Excellent — volume matters.

Customer Behaviour

Acceptance patterns, repayment outcomes, and behavioural archetypes

Customer Behaviour & Credit State Evolution



Four Behavioural Archetypes

Reliable Repayer

Exc / upper-Good

Accepts most offers, repays consistently. Stable and predictable. Retention risk: competitor offers. → Every-campaign eligibility.

Upward Mover

lower-Good / upper-Fair

Consistent improvement via on-time repayments. Highest credit-improvement ROI. → Regular offers, track migration velocity monthly.

Slow Recoverer

lower-Fair / upper-Poor

Progress but at slower pace — often recovering from one adverse event. → Every-other-campaign; monitor completion trends.

Persistent Risk

lower-Poor

Remains Poor despite offers. Low completion, no benefit from extensions. → Freeze offers; route to credit counselling.

Credit Migration

How the portfolio quality evolves over 12 months

Start → End	Excellent	Good	Fair	Poor
Excellent	99.5%	0.2%	0.2%	0.1%
Good	81.8%	17.1%	0.4%	0.7%
Fair	31.8%	52.0%	14.1%	2.0%
Poor	0.9%	30.6%	40.2%	28.3%

Good → Excellent
81.8%

The dominant upgrade pathway.
Compounding on-time boosts drive majority across threshold in 4–6 months.

Fair → Good / Excellent
83.8%

Fair is a transition state, not steady.
The programme is a credit rehabilitation pathway.

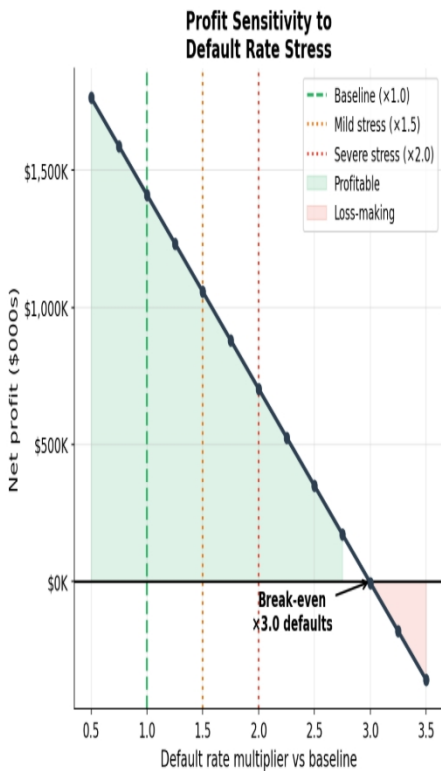
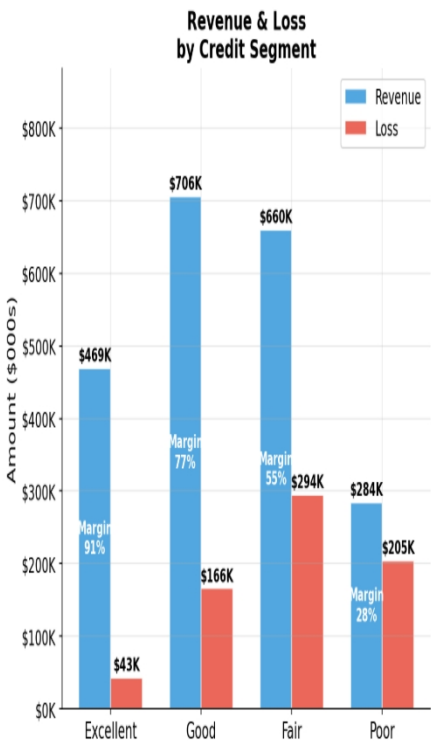
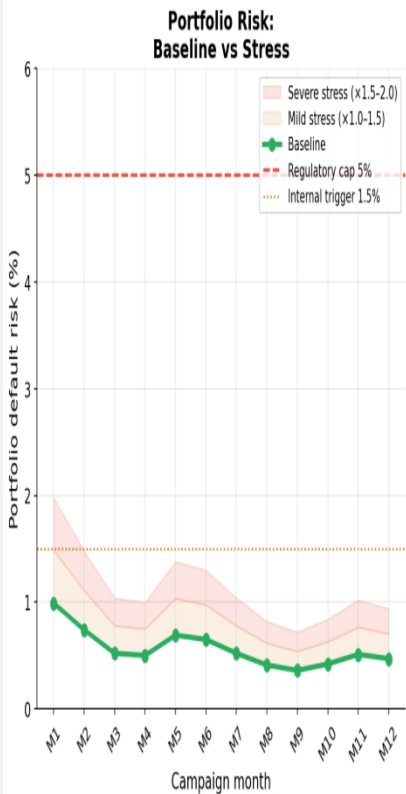
Poor: three-way split
28.3% remain Poor

30.6% → Good (recoverable)
40.2% → Fair (slow improvers)
28.3% stay Poor (freeze offers)

Risk Profile

Default rates, LP constraint utilisation, and stress resilience

Risk Analysis & Stress Testing



Risk Summary

0.66%

Portfolio Default Rate
(5% LP hard cap — 7.5× headroom)

0.99%

Peak Campaign Risk
(Month 1 — never exceeded since)

\$707K

Total Credit Losses
(\$2.12M revenue → 66.6% margin)

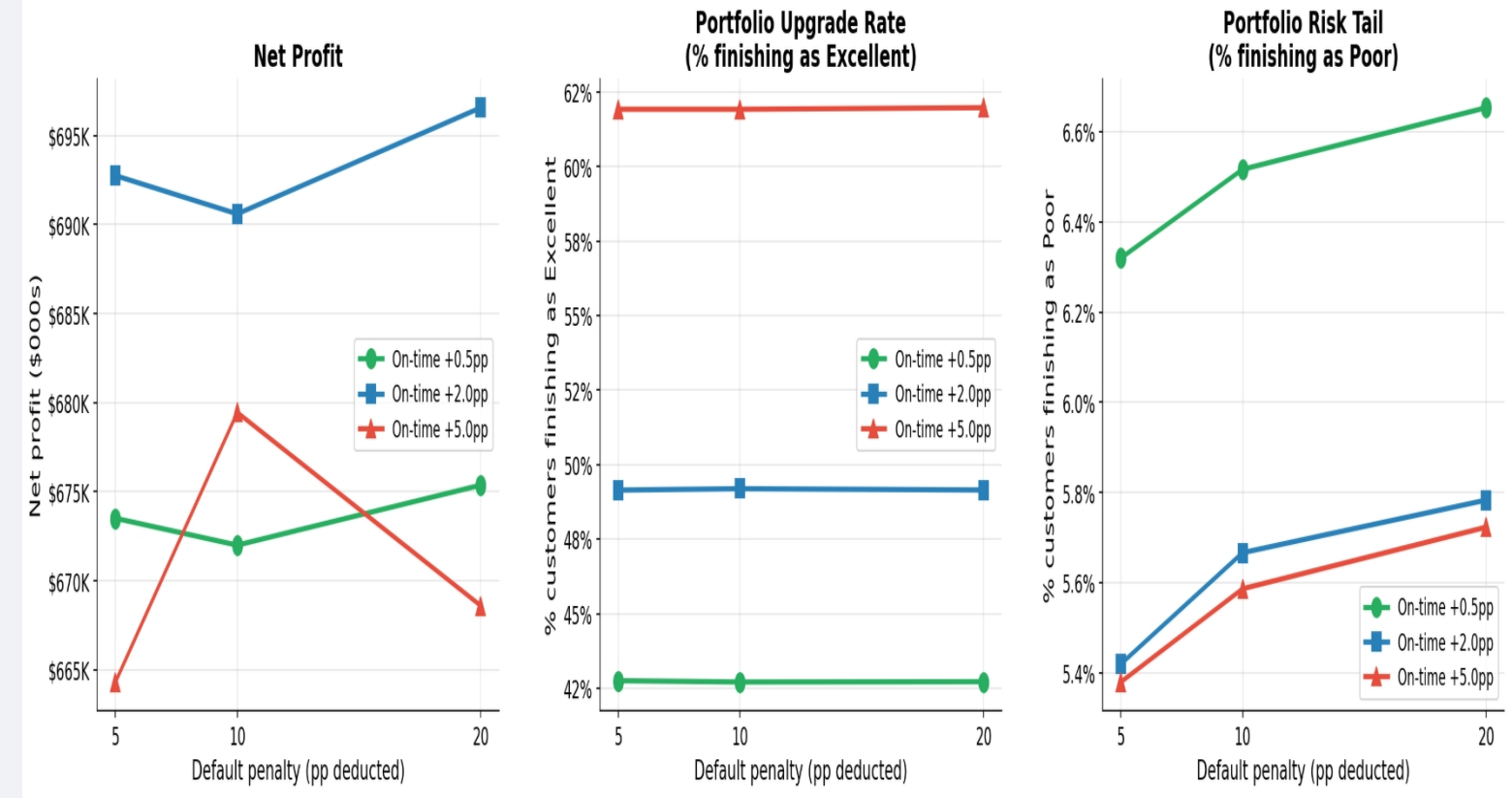
Stress Test Resilience - Net Profit

1.0×	Baseline	\$1,411K	✓ Profitable
2.0×	Moderate	\$714K	✓ Profitable
3.5×	Severe	\$26K	✓ Breakeven
5.0×	Extreme	-\$648K	✗ Loss

Sensitivity Analysis

Which parameter levers move the dial — and by how much

Payment Boost Sensitivity – 180d / Weekly Campaigns



Key Findings

On-Time Boost is the Dominant Lever

+5.0pp moves 19.2pp more customers to Excellent (43% → 62%). But profit peaks at 2.0pp — higher boosts over-promote customers, shrinking the eligible pool and reducing revenue.

Recommended: 2.0pp / 10pp

Balances profit (\$690K sub-run), quality improvement (49.2% Excellent), and controlled default rate (0.62%). This is the baseline configuration.

Default Penalty is Second-Order

Variance across 5/10/20pp settings is < 1% of total profit. Primary effect: faster exit from Poor tier. 10pp avoids cliff-edge dynamics.

Evaluation Warning : The document was created with Spire Presentation for Python

Tier Strategy & Policy Decision Matrix

Recommended offer cadence, limit sizing, and approval logic by tier

Credit Limit Extension — Policy Recommendation by Credit Tier



Tier	Offer Cadence	Limit Increase	Priority	K
Excellent	Every campaign	+20–25%	Maximum volume	
Good	Every campaign	+15–20%	Standard approval	I
Fair	Every 2nd campaign	+10–15%	Selective	
Poor	Every 3rd campaign	+5–10%	Enhanced review	C

Risk Budget Status: 0.66% realised vs 5.0% cap — 7.5× headroom.
Room to expand Good/Excellent volume without constraint violations.